

ShopSphere — Cloud Service Classification

Digital Egypt Cubs Initiative — Level 5, Task 2.2

Student ID: 30908170106047

The production deployment from Task 1 uses three cloud services. Each is classified below by service model — IaaS, PaaS, or SaaS — with a one-line justification for the classification.

Service	Classification	Reason
Frontend hosting	PaaS	Vercel builds and serves the React/Vite frontend; the student supplies application code only, while Vercel manages the build pipeline, CDN, TLS certificates, and servers.
Backend hosting	PaaS	The Express API runs as a Vercel serverless function; the student deploys application code and Vercel provisions, scales, and patches the underlying compute automatically.
Database (Supabase)	PaaS	Supabase provides a fully managed PostgreSQL instance; the student designs the schema and queries data, while Supabase handles provisioning, patching, backups, and availability.

Why not IaaS or SaaS?

Not IaaS: none of the three services expose raw virtual machines, block storage, or networking configuration to the student. There is no OS to patch and no server to provision — all infrastructure below the application layer is abstracted away, which rules out IaaS.

Not SaaS: none of the three services are ready-made end-user applications that the student simply consumes. In every case the student writes and deploys original application code (frontend UI, API routes, or a custom schema) on top of the platform, which is the defining trait of PaaS rather than SaaS.

Borderline note on Supabase: Supabase also offers SaaS-like managed features (auth, storage, dashboards), but in this project it is used purely as a hosted PostgreSQL database that the backend connects to via a connection string — the student defines and owns the schema, so it is classified here as PaaS (specifically Database-as-a-Service, a PaaS sub-category).